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StreamFlix KPI Calculations Report
Phase 3: KPI Calculations
Kindly view the Python Notebook file at this link

Data Used

Table	Rows
subscribers.csv	15,000
titles.csv	9,000
watch_history.csv	443,786
watchlist.csv	65,000

Cohort retention uses 1 June 2026 as the snapshot date.

KPI Summary Table

#	KPI	Value	Target	Status
1	Total Watch Hours	2,274,135	-	-
2	Active Rate	74.66%	> 70%	On target
3	Churn Rate	25.34%	< 30%	On target
4	Avg Completion Rate	65.33%	> 60%	On target
5	Monthly Recurring Revenue (MRR)	$175,868.51	-	-
6	ARPU	$15.70	-	-
7	Avg Watch Time / Subscriber	203.07 hrs	-	-
8	Watchlist Conversion	46.23%	> 40%	On target
9	Hit Concentration	31.10%	-	-
10	Originals Share of Hours	27.18%	-	-

KPI Details

- Total Watch Hours: sum of watch_duration_min ÷ 60 = 2,274,135 hours.
- Active Rate: 11,199 active out of 15,000 subscribers = 74.66%. Above the 70% target.
- Churn Rate: 3,801 inactive out of 15,000 = 25.34%. Below the 30% limit.
- Avg Completion Rate: average of completion_pct = 65.33%. Above the 60% target.
- MRR: sum of monthly price for active subscribers = $175,868.51.
- ARPU: MRR ÷ 11,199 active subscribers = $15.70 per month.
- Avg Watch Time / Subscriber: total watch hours ÷ active subscribers = 203.07 hours
- Watchlist Conversion: 30,048 watched out of 65,000 entries = 46.23%. Above the 40% target.
- Hit Concentration: the top 10% of titles (900 of 9,000) get 31.10% of all plays.
- Originals Share of Hours: 618,153 hours on Originals = 27.18% of all hours.

Bonus: Cohort Retention

Subscribers are grouped by signup month. A subscriber is counted as retained at 3, 6, or 12 months if they stayed at least that long; only subscribers who signed up at least that many months before the snapshot date are counted.

Overall retention:

Time after signup	Retained
3 months	98.28%
6 months	95.72%
12 months	92.46%

Retention by signup year:

Signup year	3-month	6-month	12-month
2016	99.16%	97.89%	96.62%
2017	99.65%	99.12%	97.36%
2018	99.28%	98.44%	96.29%
2019	99.24%	97.91%	96.02%
2020	99.11%	97.78%	95.33%
2021	99.31%	97.54%	94.44%
2022	98.73%	96.95%	93.72%
2023	99.07%	96.50%	91.62%
2024	98.53%	95.26%	89.10%
2025	96.26%	88.87%	81.02%
2026	89.88%	-	-

Key Findings

- Targets met: Active Rate, Churn Rate, Avg Completion Rate, and Watchlist Conversion are all on target.
- Revenue: MRR is $175,868.51, with ARPU of $15.70 per active subscriber.
- Hits: The top 10% of titles drive 31.10% of plays.
- Originals: Originals make up 27.18% of watch hours.
- Retention is falling for newer cohorts: 12-month retention drops from 97.36% (2017 signups) to 81.02% (2025 signups). 2025 and 2026 cohorts also lose more subscribers in the first 3 and 6 months.

Note: Churn Rate here counts all inactive subscribers out of all subscribers, as defined in the brief. It is not a monthly churn rate.